Super Resolution Guided Deep Network for Land Cover Classification From Remote Sensing Images

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***Abstract*— The low resolution of remote sensing images often limits the land cover classification (LCC) performance. Super resolution (SR) can improve the image resolution, while greatly increasing the computational burden for the LCC due to the larger size of the input image. In this article, the SR-guided deep network (SRGDN) framework is proposed, which can generate meaningful structures from higher resolution images to improve the LCC performance without consuming more computational costs. In general, the SRGDN consists of two branches (i.e., SR branch and LCC branch) and a guidance module. The SR branch aims to increase the resolution of remote sensing images. Since high- and low-resolution image pairs cannot be directly provided by imaging sensors to train the SR branch, we introduce a self-supervised generative adversarial network (GAN) to estimate the downsampling kernel that can produce these image pairs. The LCC branch adopts the high-resolution network (HRNet) to retain as much resolution information with a few downsampling operations as possible. The guidance module teaches the LCC branch to learn the high-resolution information from the SR branch without the utilization of the higher-resolution images as the inputs. Furthermore, the guidance module introduces spatial pyramid pooling (SPP) to match the feature maps of different sizes in the two branches. In the testing stage, the guidance module and SR branch can be removed, and therefore do not create additional computational costs. Experimental results on three real datasets demonstrate the superiority of the proposed method over several well-known LCC approaches.**

***Index Terms*— Guidance, land cover classification (LCC), remote sensing image, super resolution (SR).**

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I. Introduction

LAND cover classification (LCC) plays an important role in many applications, such as traffic flow prediction [1], ecological management [2], and urban planning [3]–[5]. With the rapid development of remote sensing technology, a large number of remote sensing images are available for the LCC. Under this circumstance, the efficient LCC from remote sensing images has attracted considerable attention of researchers.

The LCC is to assign each pixel in the remote sensing image with a predefined class. In the past decades, several classification methods [6]–[8] are proposed by exploiting the characteristics of the predefined class for some applications. For example, the road connectivity is usually used for road extraction [1], [9], which is essential for urban traffic prediction. In [2] and [10], utilizing the difference between red radiation and near-infrared radiation to extract vegetation, which is useful for ecological management, is a classical strategy. Moreover, in the urban planning task, road extraction, building extraction, water extraction, and vegetation extraction are required to comprehensively decide how to arrange commercial areas, residential areas, and parks [3], [11]. However, various imaging condition and geographical areas lead to high diversity for the same class and low separability among different classes [12]. This makes it difficult to represent the same class of land cover and distinguish different classes of land cover.

In order to effectively represent and distinguish land cover from remote sensing images, the spatial information and the spectral information are utilized based on the past decades of expert experience. In Gonzalez-Alonso *et al.* [13] improve the LCC results from two remote sensing images in central Spain by taking advantage of spatial contextual information. In [3], some spectral indices [e.g., the normalized difference built-up index (NDBI), the normalized difference water index (NDWI), the modified NDWI (MNDWI), the normalized difference vegetation index (NDVI), and the bare soil index (BI)] are used to

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generate initial samples for the iteratively active learning. With the deepening of the knowledge of experts, both spatial and spectral information has been researched in some LCC studies. Fang *et al.* [14] utilized superpixel approach and multiple kernels strategy to exploit spatial and spectral information, respectively, which can achieve satisfactory LCC results for some hyperspectral images. In [15], the edge-preserving features filter is used to extract spatial-spectral features and the principal component analysis method highlights the spectral separability of features.

All the above methods rely heavily on prior information provided by experts. However, remote sensing images contain rich details of the interest land cover, including geometrical shapes and structural content, which is difficult to be covered by expert experience. A typical way for addressing this problem is to introduce statistical prior information into the LCC [16]–[21]. The support vector machine (SVM) and multinomial logistic regression (MLR) are two kinds of popular classifiers used after the feature extraction process for the LCC from remote sensing images [16]–[18]. In [19], a quasi-exhaustive feature bank, including NDVI and texture feature, is provided first and a boosting classifier is designed to select optimal features for the final LCC. To jointly utilize unlabeled samples and training samples, the expectation–maximization algorithm is employed to estimate the conditional probability density functions of the unknown class and the classes of interest, and then the Markov random fields are adopted to improve prediction accuracies in [20]. Moreover, in [21], the optimum-path forest is proposed for the LCC based on the distributions of the classes of interest, which is suitable for both unsupervised and supervised situations. These methods mainly adopt the hand-crafted features and thus the performance of the LCC from remote sensing images is limited.

To solve this problem, deep learning is introduced to automatically extract representative and discriminative features, and achieve high-precision classification results. It consists of many layers for the feature extractor and a nonlinear classifier, whose parameters are trained by the backward propagation algorithm. Convolutional neural network (CNN), one of the most famous deep learning framework, has shown its powerful strength in many fields, including images classification [12], super resolution (SR) [22], and semantic segmentation [23]. Recently, the CNN model has been extended to the LCC and achieved excellent classification results. In [24], it is found that the pixel-to-pixel method is superior to the patch-based method in the LCC through experiments on a real remote sensing dataset. Sang *et al.* [23] applied the attention mechanism, including the self-attention mechanism and the fusion attention mechanism, to enhance the LCC results. In [25], to handle the problem of domain adaptation, a generative adversarial network (GAN) is added into a CNN model for the LCC from multiple-source remote sensing images. Unlike nature images, the resolution of remote sensing images is not very high, which limits the performance of all these approaches based on the CNN model.

To break through the limitation of the resolution of remote sensing images for the LCC, the SR is an effective way. Similar to the LCC methods, the state-of-the-art SR methods are based on the CNN model. In [26], the CNN model is first proposed for the SR, which greatly enhances the results of the SR. In [27], the second-order multiscale information is introduced into the CNN model to improve the SR performance. In the [28], the residual idea is fully utilized to design the CNN model, which exhibits very fine edges and textures. Lei *et al.* [29] proposed a framework to simultaneously obtain SR and LCC results from remote sensing images. The above method usually requires high- and low-resolution image pairs to train the CNN model, but the high-resolution images cannot be directly provided by imaging sensors. In general, high-resolution images are original images, while low-resolution images are downsampled from original images by some interpolation methods (such as the nearest neighbor, bilinear, and bicubic) [30]–[32]. These fixed downscaling kernels cannot reflect the real downsampling process. In addition, the SR method produces input images with a larger size for the LCC, which leads to more computational cost.

To address the above issues, in this article, a new SR-guided deep network (SRGDN) architecture is designed. This architecture can generate fine structures without more computational cost. The SR branch, the LCC branch, and a guidance module constitute the architecture of the SRGDN. The SR branch takes the enhanced deep residual (EDSR) networks as the basic framework. The downscaling kernels of each remote sensing image are estimated by exploiting its own multiscale structures, which can generate effective high- and low-resolution image pairs to train the EDSR model [33]. In addition, the high-resolution network (HRNet) [34] is taken as the backbone of the LCC branch since it can preserve much resolution information. The SR branch and the LCC branch share some feature extractors to reduce parameters. The guidance module allows feature maps in the SR branch to guide another branch to produce feature maps with finer details. In detail, the cosine distance is used to obtain self-similarity matrix of feature maps in two branches and the $l1$ norm is used to measure the distance of two self-similarity matrices. Since the sizes of the two matrices do not match, the guidance module introduces the spatial pyramid pooling (SPP) [35] to extract two representative maps of the same size from global to local by a fixed number of spatial bins. It can be observed that the size of input images for the LCC branch does not increase, which reduces training times. Furthermore, the SR branch and the guidance module can be removed at the testing phase, thereby reducing computational cost of testing. The results of the experimental part prove the effectiveness of the SRGDN architecture.

The main contributions of this study are summarized below.

1) A novel end-to-end deep network, including an LCC branch, an SR branch, and a guidance module, is proposed to produce more detailed structures for the LCC results under the guidance of the SR images without more computational costs.
2) The self-similarity matrix based on cosine distance is proposed to establish the relationship between the feature maps of LCC and SR, and the SPP is introduced to match the size of the self-similarity matrix from the SR branch and the LCC branch.

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3) A self-supervised GAN is utilized to generate high- and low-resolution image pairs, which cannot be directly obtained from imaging sensors, to train the SR branch.

The remainder of this article is organized as follows. Section II reviews the related works of the HRNet and the EDSR. Section III introduces the proposed SRGDN for the LCC from remote sensing images. Section IV shows our experimental results, conducted on three datasets. Section V concludes this article and hints at plausible future research directions.

II. Related Works

A. CNN-Based LCC Methods

CNNs have been demonstrated to be a powerful tool in the LCC task [23]–[25]. Existing state-of-the-art CNN-based methods for the LCC from remote sensing images are based on the convolutional encoder–decoder framework. Specifically, remote sensing images are first encoded as a low-resolution representation. A pretrained CNN model, such as ResNet [36] and DenseNet [37], is used to encode remote sensing images. Then, this low-resolution representation is decoded into high-resolution results by some deconvolutional layers. The resolution information has been lost in the encoding stage, and to recover the resolution information, many studies have focused on how to enhance the decoding results [38]–[40]. However, the recovery from low-resolution representation to high-resolution result is an ill-posed problem. Encoding remote sensing images without reducing resolution information is an effective way to avoid this problem in the CNN-based LCC method.

The HRNet is a representative method that can encode input images while preserving resolution information [34]. The HRNet encodes input images by different resolution streams in parallel. The highest-resolution information is maintained in the first stream and high-to-low resolution streams are added stage by stage. The HRNet consists of several stages and the nth stage contains n streams. At the first stage, the HRNet is started from the first stream. The later stages contain two kinds of streams, including the resolutions from the previous stage and the extra lower resolutions through downsampling, respectively. The information of each resolution stream is coded once at each stage. Each encoding contains several blocks and each block contains a convolution operation, a batch normalization (BN) operation, and a nonlinear transformation operation. Since the encoding process does not lose resolution information from downsampling process, the first stream can retain all resolution information. Furthermore, multiresolution information fusion is a classic and effective strategy for the LCC tasks [41]–[43]. After each encoding process from the second stage to the final stage, this strategy is repeatedly used $n - 1$ times. Multiresolution repeated fusion enhances representation capabilities by exchanging information across different resolution streams. Finally, feature maps from all streams are concatenated to make land cover predictions.

B. CNN-Based SR Methods for Remote Sensing Images

Due to the limited resolution of remote sensing images, it is difficult to extract some fine details. The SR methods can effectively reconstruct the high-resolution images from low-resolution images. The SR methods for remote sensing images based on supervised learning, especially CNN model, have excellent performance [27], [28]. SR from low-resolution images to high-resolution images is also an ill-posed information. These methods need high- and low-resolution image pairs as prior information to estimate their mapping relationship.

In [26], the CNN method is first introduced into the SR task and exhibits powerful performance on the SR task. To obtain better results, a deeper structure is an effective strategy. The residual module is widely used in the deep CNN [44]–[46] because it helps to converge to the optimal solution [36]. However, the residual module is proposed for high-level computer vision tasks such as image classification, segmentation, and detection, and is suboptimal for low-level computer vision tasks such as image SR. To solve this problem, the EDSR is proposed in [28]. Specifically, the EDSR is similar to [45], but the BN layers are totally removed, which not only retains range flexibility from networks but also reduces memory usage of graphics processing unit (GPU) during training. Then rectified linear unit (ReLU) layers outside the residual module are removed in the EDSR to widen the mapping range. In addition, the original image is input without upsampling and the pixel shuffle layer [47] is utilized to upscale feature maps at the final network. Finally, the $l1$ norm is adopted as the loss function.

III. Proposed Method for LCC From Remote Sensing Images

The CNN-based methods can obtain the state-of-the-art LCC performance from remote sensing images, but the limited resolution of remote sensing images is still the main obstacle for the performance of these methods. The SR methods, especially based on CNN, can improve the resolution of remote sensing images and help the LCC methods to acquire better results [29]. However, SR methods based on CNN need high- and low-resolution image pairs, where high-resolution images are not obtainable from imaging devices, to train their own parameters. Furthermore, images with higher resolution result in more computational cost.

To address these issues, we design a novel end-to-end CNN framework, i.e., SRGDN, which consists of an LCC branch, an SR branch, and a guidance module. The SR branch guides the LCC branch to capture information with more detailed structures by the guidance module. Moreover, this framework efficiently improves computational efficiency in the training and testing phases. In addition, to train the SR branch, a GAN is utilized to generate high- and low-resolution image pairs through estimating the downsampling kernel. In this section, the SRGDN and the GAN-based downsampling kernel estimation are described in detail below.

A. Architecture of the SRGDN

The architecture of the proposed SRGDN is illustrated in Fig. 1. The LCC branch, the SR branch, and the guidance module constitute this framework. The LCC branch and the SR branch are based on the HRNet and the EDSR, respectively. The SR branch is used to increase the resolution of remote

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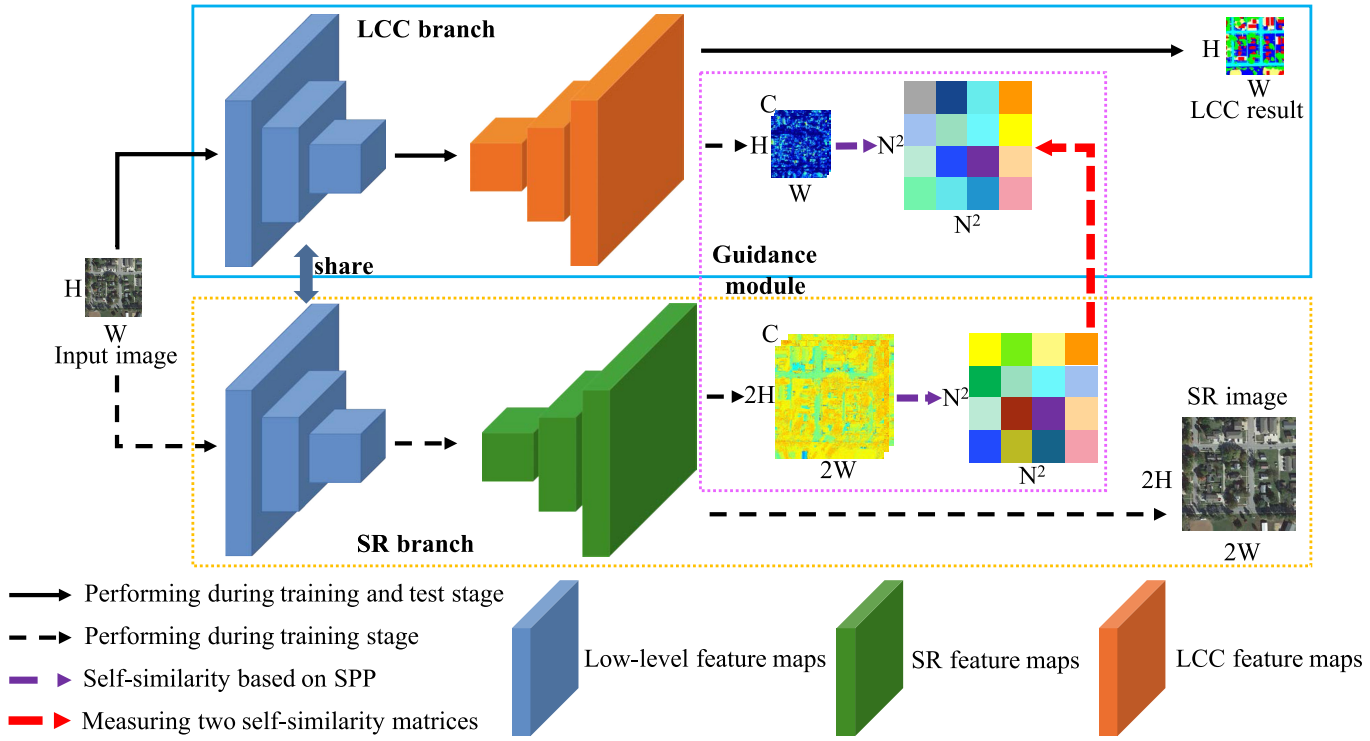


Fig. 1. Proposed framework (SRGDN) for the LCC from remote sensing images.

sensing images, which helps the LCC branch to better predict the LCC results from remote sensing images. Furthermore, in order to reduce parameters, the LCC branch and the SR branch share some low-level layers.

1) LCC Branch: The size of the LCC branch normalized output $\boldsymbol{Y^{LCC}}$ $(0 \leqslant \boldsymbol{Y^{LCC}} \leqslant 1)$ is $W \times H \times K$, where W and H are the width and height of the input image, respectively, and K is the number of classes. The cross entropy loss function for the LCC branch is denoted as

$$L^{\mathrm{LCC}} = -\frac{\sum_{i=1}^{W}\sum_{j=1}^{H}\theta_k \cdot \log\left(Y_{i,j}^{\mathrm{LCC}}\right)}{WH} \tag{1}$$

where $Y_{i,j}^{\mathrm{LCC}} = \{Y_{i,j,1}^{\mathrm{LCC}}, \ldots, Y_{i,j,K}^{\mathrm{LCC}}\}$ $(\sum_{k=1}^{K} Y_{i,j,k}^{\mathrm{LCC}} = 1)$ and the θ_k represents a vector with 1 for the true label and 0 entries for the others.

2) SR Branch: The size of the SR branch output $\boldsymbol{Y^{SR}}$ $(0 \leqslant \boldsymbol{Y^{SR}} \leqslant 255)$ is $2W \times 2H \times 3$ (the upscaling factor is 2). The $l1$ loss function for the SR branch can be obtained by

$$L^{\mathrm{SR}} = -\frac{\left\|Y^{\mathrm{SR}} - Y^{\mathrm{HR}}\right\|_1}{4WH} \tag{2}$$

where the Y^{HR} is the high-resolution image with a shape of $2W \times 2H \times 3$. The Y^{HR} is the ground truth for the SR branch.

3) Guidance Module: If the SR images are directly input to the LCC branch, they would cause too much computational cost. Therefore, a guidance module is designed to allow the feature maps in the LCC branch to learn detailed information from the feature maps in the SR branch without the more computational cost of the LCC branch. The feature maps at the final SR branch are selected to guide the feature maps at the final LCC branch. Moreover, due to the discrepancy of feature distribution between the SR branch and the LCC branch, a transfer block, including an 1×1 convolution layer, a BN layer, and a ReLU layer, is appended after the SR branch [48]. The spatial dimensions of the final feature maps in the SR branch and the LCC branch are $2W \times 2H$ and $W \times H$, respectively. As a matter of fact, it is not necessary to fix the upscaling factor to 2 and any upscaling multiple can be adopted. Thus, the relationship between their spatial dimensions is not matched and fixed, and it is difficult to design a guidance relationship between them. To solve this problem, the SPP layer is used to obtain fixed-size feature maps [35]. From the high pyramid level to the low pyramid level, the information reflected by the feature vector ranges from coarse to fine. The first pyramid level covers the entire remote sensing image, and it reflects the global information. Assume that the size of the feature maps $\boldsymbol{F} = \{F_1, \ldots, F_N\}$ from the SPP layer is $C \times N$, where C is the length of each feature vector and N is the number of feature vector. In this case, although feature maps with different resolutions from the two branches are input in the SPP layer, the sizes of the output feature maps are the same and the corresponding feature vector represents the information of the same region. Due to the large difference between feature maps from the SR branch and the LCC branch, even though the transfer block is introduced, the feature maps from the SR branch cannot be utilized to directly guide the feature maps from the LCC branch. For these feature maps from the two branches, the similarity between each two feature vectors is obtained based on the cosine distance. This similarity can be expressed

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as a self-similarity matrix, which is defined as

$$S_{i,j} = \left(\frac{F_i}{\|F_i\|_2}\right)^T \cdot \left(\frac{F_j}{\|F_j\|_2}\right) \tag{3}$$

where $1 \leqslant i \leqslant N$ and $1 \leqslant j \leqslant N$. The $\boldsymbol{S^{SR}} = \{S^{\mathrm{SR}}_{1,1}, \ldots, S^{\mathrm{SR}}_{N,N}\}$ and the $\boldsymbol{S^{LCC}} = \{S^{\mathrm{LCC}}_{1,1}, \ldots, S^{\mathrm{LCC}}_{N,N}\}$ represent the self-similarity matrix of the SR branch and the LCC branch, respectively. The larger the $S_{i,j}$ ($0 \leqslant S_{i,j} \leqslant 1$), the higher the similarity between the two regions. The similarity of the same class of land cover is high, and thus the two self-similarity matrices should tend to be similar. The $l1$ norm is utilized to constrain this similar, which is defined as

$$L^{\mathrm{similar}} = \frac{\|\boldsymbol{S^{SR}} - \boldsymbol{S^{LCC}}\|_1}{N \times N}. \tag{4}$$

4) Training and Testing: Based on the above descriptions, the final loss function of the SRGDN is defined as

$$L = L^{\mathrm{LCC}} + 0.1L^{\mathrm{SR}} + L^{\mathrm{similar}}. \tag{5}$$

During the training time, the SRGDN is optimized end-to-end. To minimize the L, the stochastic gradient descent (SGD) algorithm is generally applied to achieve this goal. In addition, the backward propagation algorithm is used to update all parameters of the SRGDN. Through hundreds of iterations, all parameters of the SRGDN tend to be optimal. Compared with that directly inputs the remote sensing images from EDSR to HRNet, the input size of the LCC branch in SRGDN does not increase without creating higher computational costs in the training phase. During the testing time, the SR branch and the guidance module can be removed since the parameters of the LCC branch have been fixed, and thereby the computational cost of the SRGDN is the same as the HRNet.

B. GAN-Based Downsampling Kernel Estimation

Although the EDSR can output satisfactory high-resolution images, this model requires high- and low-resolution image pairs for training. Limited by remote sensing sensors, high-resolution remote sensing images cannot be directly obtained. In general, original images are downsampled by a kernel function (such as the nearest neighbor, bilinear, and bicubic) to produce low-resolution images [26]–[29]. Then, the original images and the low-resolution images constitute high- and low-resolution image pairs. Nevertheless, these specific downsampling kernel functions cannot reflect the real mapping relationship between the high-resolution images and the corresponding low-resolution images. This real mapping relationship depends on the imaging condition, and it is difficult to design a general model to describe this relationship. For a high-resolution image and its corresponding low-resolution image, their pixel distributions are very similar. In addition, multiscale information, such as buildings with different sizes, exists in a remote sensing image, which can be considered as a manifestation of multiresolution information. Thus, a self-supervised GAN is proposed, shown in Fig. 2, to use the real mapping relationship and the similar data distributions between the high-resolution image and its corresponding low-resolution image for adversarial training [33]. The GAN for

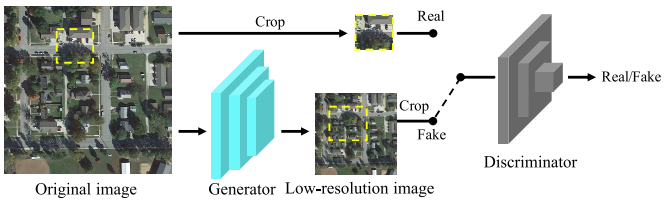


Fig. 2. GAN-based downsampling kernel estimation for generating low-resolution remote sensing image.

downsampling kernel estimation consists of a generator and a discriminator, where the generator learns the mapping relationship and the discriminator distinguishes whether the data distribution between the high-resolution image and the corresponding low-resolution image is similar. The corresponding low-resolution image is generated from the original image through this mapping relationship, and the original image and the corresponding low-resolution image are used to train the SR branch of the SRGDN.

The remote sensing image is first cropped into patches of the size $M \times M$ and $2M \times 2M$. The patches X^{HR}_{2M} of the size $2M \times 2M$ are input into the generator to produce low-resolution patches X^{LR}_{M} of the size $M \times M$. These low-resolution patches X^{LR}_{M} are input into the discriminator together with the patches X^{HR}_{M} of the size $M \times M$ cropped from the remote sensing image. Then, the discriminator and the generator are optimized alternately.

Specifically, the parameters of the generator are frozen and the discriminator is first optimized. The discriminator for patches is based on a fully convolutional network (FCN) [49]. The parameters and the output of the discriminator are defined as $\boldsymbol{D}$ and $\boldsymbol{D}(\boldsymbol{X})$ ($0 \leqslant \boldsymbol{D}(\boldsymbol{X}) \leqslant 1$), respectively. The objective function of the discriminator is denoted as

$$\boldsymbol{D} = \mathrm{argmax}\left(\left\|\boldsymbol{D}\left(X^{\mathrm{HR}}_{M}\right)\right\|_1 + \left\|\boldsymbol{D}\left(X^{\mathrm{LR}}_{M}\right) - 1\right\|_1\right) \tag{6}$$

where the output of the X^{LR}_{M} and X^{HR}_{M} are hoped to discriminate between fake and true, i.e., 0 and 1, respectively.

Then, the parameters of the discriminator are frozen and the generator is optimized. Since the real down-sampling process is a linear mapping relationship [50]–[52], the linear generator, composed of some convolution layers, is adopted. In this case, all local minima are equivalent to the same solution, because all mapping relationships in the generator are linear [53]. Note that, compared with the network with a single linear mapping layer, the network with multiple linear mapping layers is easier to be optimized [54]. Furthermore, the convolution layers are initialized to be approximately equivalent to a downsampling kernel function (bicubic in this article). Assume that P convolution kernels $G^p = \{G^p_{1,1,1}, \ldots, G^p_{c,h,b}, \ldots, G^p_{C,H,B}\}$ with size of $C \times H \times B$ exist in the generator. The parameters of the generator are defined as $\boldsymbol{G}$ and the objective function of the generator is denoted as

$$\boldsymbol{G} = \mathrm{argmin}\left(\left\|\boldsymbol{D}\left(X^{\mathrm{HR}}_{M}\right)\right\|_1 + \left\|\boldsymbol{D}\left(X^{\mathrm{LR}}_{M}\right) - 1\right\|_1 + 0.5\left(\sum_{p=1}^{P}\left\|\sum_{b=1}^{B}\sum_{h=1}^{H}\sum_{c=1}^{C} G^p_{c,h,b} - 1\right\|_1\right)\right.$$

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$$+0.5\left(\sum_{p=1}^{P}\left\|\sum_{b=1}^{B}\sum_{h=1}^{H}\sum_{c=1}^{C}G_{c,h,b}^{p}\cdot R_{c,h,b}\right\|_{1}\right)+5\left(\sum_{p=1}^{P}\sum_{b=1}^{B}\sum_{h=1}^{H}\sum_{c=1}^{C}\left\|G_{c,h,b}^{p}\right\|_{1/2}\right)+\left(\sum_{p=1}^{P}\left\|\left(\frac{C+1}{2},\frac{H+1}{2},\frac{B+1}{2}\right)-\frac{(c,h,b)\sum_{b=1}^{B}\sum_{h=1}^{H}\sum_{c=1}^{C}G_{c,h,b}^{p}}{\sum_{b=1}^{B}\sum_{h=1}^{H}\sum_{c=1}^{C}G_{c,h,b}^{p}}\right\|_{2}\right)\right) \quad (7)$$

where $R_{c,h,b}$ is the weight that increases exponentially with the distance from the center $(((C+1)/2), ((H+1)/2), ((B+1)/2))$. The objective function includes four regularization terms about convolution kernels. The first regularization term $\sum_{p=1}^{P}\|\sum_{b=1}^{B}\sum_{h=1}^{H}\sum_{c=1}^{C}G_{c,h,b}^{p}-1\|_1$ is to encourage the sum of a convolution kernel to be 1. The second regularization term $\sum_{p=1}^{P}\|\sum_{b=1}^{B}\sum_{h=1}^{H}\sum_{c=1}^{C}G_{c,h,b}^{p}\cdot R_{c,h,b}\|_1$ is to penalize nonzero values around the boundary. The third regularization term $\sum_{p=1}^{P}\sum_{b=1}^{B}\sum_{h=1}^{H}\sum_{c=1}^{C}\|G_{c,h,b}^{p}\|_{1/2}$ is a sparse constraint. The last regularization term $\sum_{p=1}^{P}\|(((C+1)/2),((H+1)/2),((B+1)/2))-((c,h,b)\sum_{b=1}^{B}\sum_{h=1}^{H}\sum_{c=1}^{C}G_{c,h,b}^{p})/(\sum_{b=1}^{B}\sum_{h=1}^{H}\sum_{c=1}^{C}G_{c,h,b}^{p}))\|_2))$ is to encourage the center of the mass to be the center of the convolution kernel. The discriminator and the generator are optimized in turn.

IV. Experimental Results

A. Datasets Description

To verify the effectiveness of the proposed SRGDN, A set of comprehensive experiments are performed on three remote sensing datasets. The first one is obtained from the sensor of unmanned airborne vehicle (UAV), the second one is obtained from the sensor of Gaofen-2 satellite, and the rest one is a publicly benchmark dataset, i.e., Potsdam dataset.

1) The UAV image dataset consists of 10 pixel-level annotated images divided into the background and seven classes of interest, including building, tree, car, sparse grass, grass, road, and parking lot. All images were distributed in the third "Geovis Technology Cup" High-Resolution Image Interpretation Software Competition. These images have a spatial size of 2000 × 2000, and the spatial resolution of these images is approximately 0.15 m per pixel. Fig. 3 shows an example of a remote sensing image in the UAV image dataset.
2) The Gaofen-2 satellite image dataset consists of 10 pixel-level annotated images divided into the background and three classes of interest, including water, vegetation, and building. All images were distributed in the 2019 Remote Sensing Image Sparse Representation and Intelligence Analysis Contest. These images were taken from 2015 to 2016 over city, town, forest, mountain, and so on. They have the spatial size of 7200×6800 with a spatial resolution of 4 m. Fig. 4 shows an example of the Gaofen-2 satellite image dataset.

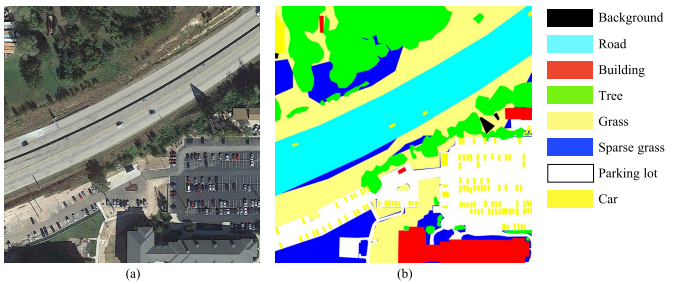


Fig. 3. Example of remote sensing image and its ground truth in the UAV image dataset. (a) Original image. (b) Ground truth.

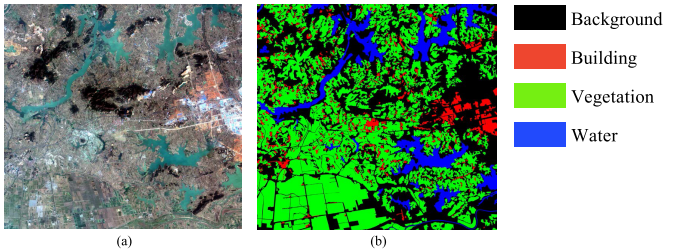


Fig. 4. Example of remote sensing image and its ground truth in the Gaofen-2 satellite image dataset. (a) Original image. (b) Ground truth.

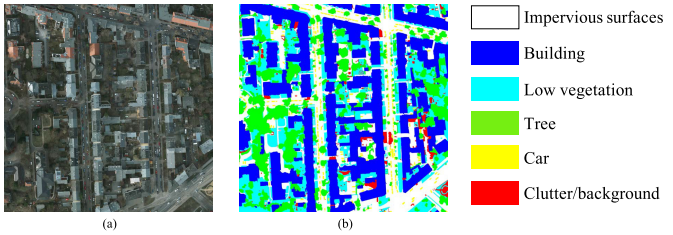


Fig. 5. Example of remote sensing image and its ground truth in the Potsdam dataset. (a) Original image. (b) Ground truth.

3) The Potsdam dataset was collected by airborne sensors over Potsdam and distributed in the Woking Groups of International Journal of Photogrammetry and Remote Sensing Commission II/4 2-D semantic labeling contest (https://www2.isprs.org/commissions/comm2/wg4/benchmark/2d-sem-label-potsdam/). The dataset consists of 38 pixel-level annotated images divided into five classes of interest, including impervious surfaces, building, low vegetation, tree, and car. According to the contest, the cluster/background is not taken into consideration. These images have a spatial size of 6000 × 6000, and each pixel has a spatial resolution of 0.05 m. Fig. 5 shows an example of a remote sensing image in the Potsdam dataset.

B. Experimental Setup

To evaluate the effectiveness of the proposed SRGDN, the class accuracy (CA), over accuracy (OA), Intersection over Union (IoU), and mean IoU (mIoU) are introduced to measure all results. The full definitions of these quantitative metrics are as follows.

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TABLE I

LCC RESULTS BY THE FCN [55], PSPNET [56], HRNET [34], INPUTTING BICUBIC IMAGES INTO HRNET, INPUTTING SR IMAGES INTO HRNET AND SRGDN ON THE UAV IMAGE DATASET

Methods	FCN	PSPNet	HRNet	Bicubic images+HRNet	SR images+HRNet	SRGDN
training times	33538 seconds	7403 seconds	11471 seconds	69793 seconds	81885 seconds	25586 seconds
testing times	272 seconds	188 seconds	47 seconds	153 seconds	171 seconds	47 seconds
OA	78.52	81.87	82.56	82.91	84.12	83.81
CA of Background	27.25	71.38	68.22	96.31	97.03	84.85
CA of Road	74.88	80.57	80.23	80.38	81.54	80.73
CA of Building	88.01	91.36	90.91	90.06	91.55	91.66
CA of Tree	87.07	91.91	89.94	90.49	90.78	90.99
CA of Grass	81.35	80.09	86.73	84.92	85.62	89.19
CA of Sparse grass	50.05	64.41	54.41	59.78	61.55	55.32
CA of Parking lot	56.27	57.11	68.24	65.33	66.24	72.61
CA of car	69.88	64.53	69.63	74.08	74.51	72.21
mIoU	51.83	62.12	63.74	65.78	67.63	66.78
IoU of Background	10.06	55.53	53.73	62.89	66.75	66.16
IoU of Road	68.44	72.50	74.50	73.88	74.11	76.19
IoU of Building	74.28	79.35	83.77	84.01	84.88	84.87
IoU of Tree	77.05	80.31	78.45	80.00	80.19	78.76
IoU of Grass	64.31	68.38	69.67	71.34	72.77	70.36
IoU of Sparse grass	43.10	51.10	48.17	52.36	52.96	50.94
IoU of Parking lot	30.21	35.19	39.23	34.98	40.25	44.24
IoU of car	47.19	54.57	62.39	66.83	69.11	62.75

Let $\boldsymbol{Num^g} = \{\mathrm{Num}_0^g, \ldots, \mathrm{Num}_k^g, \ldots, \mathrm{Num}_K^g\}$ be the numbers of pixels per class in the ground truth, where the class 0 represents the background class and the K is the number of classes. Assume that the prediction results are $\boldsymbol{Num^p} = \{\mathrm{Num}_0^p, \ldots, \mathrm{Num}_k^p, \ldots, \mathrm{Num}_K^p\}$ and the correct prediction results are $\boldsymbol{Num^c} = \{\mathrm{Num}_0^c, \ldots, \mathrm{Num}_k^c, \ldots, \mathrm{Num}_K^c\}$. The CA of the kth class is defined as

$$\mathrm{CA}_k = \frac{\mathrm{Num}_k^c}{\mathrm{Num}_k^g}. \tag{8}$$

The OA is defined as

$$\mathrm{OA} = \frac{\boldsymbol{Num^c}}{\boldsymbol{Num^g}}. \tag{9}$$

The IoU of the kth class is defined as

$$\mathrm{IoU}_k = \frac{\mathrm{Num}_k^c}{\mathrm{Num}_k^p + \mathrm{Num}_k^g - \mathrm{Num}_k^c}. \tag{10}$$

The mIoU is defined as:

$$\mathrm{mIoU} = \frac{\sum_{i=1}^{K} \mathrm{IoU}_i}{K}. \tag{11}$$

The division of the training set and the testing set is fixed. According to the 2019 Remote Sensing Image Sparse Representation and Intelligence Analysis Contest, the third "Geovis Technology Cup" High-Resolution Image Interpretation Software Competition and the Woking Groups of International Journal of Photogrammetry and Remote Sensing Commission II/4 2-D semantic labeling contest, 8 images, 9 images, and 24 images are selected in the training set for the Gaofen-2 satellite image dataset, the UAV image dataset, and the Potsdam dataset, respectively. For HRNet, the more complex the model is, the better the performance is. From [34], the HRNet with 48 convolutional kernels in the initial stage (i.e., HRNet-W48) is chosen because of a tradeoff between performance and speed, and the online hard example mining is used to deal with the imbalance samples among different classes. In addition, all our experimental results are obtained as the average of ten repeated experiments. Our experiments are conducted on a PC with CPU i7-7700K, 64 GB of RAM, and two GPU (GTX 1080 Ti). The code is released at https://github.com/Aaromxj.

C. Effectiveness of the SR Method

The SR method can recover some fine structures for remote sensing images, which is helpful to the LCC results. To test and verify the effectiveness of the SR method (i.e., EDSR [28] in this article), in this section, remote sensing images after the SR method are input into the HRNet, compared with original remote sensing images. Compared with the Gaofen-2 satellite image dataset, the UAV image dataset is more suitable for this experiment because more fine structures, such as road and car, exit. Furthermore, some interpolation methods (such as the nearest neighbor, bilinear, and bicubic) can also increase the spatial size of images, but these methods cannot provide extra resolution information for images. To demonstrate it, the remote sensing images after the bicubic method are added into experiment, which is also input into the HRNet. As can be observed from Table I, the best performance is obtained by inputting SR images. This indicates that image enhancement through SR is very useful for the LCC. Compared with that directly inputs the original images, the performances obtained by inputting bicubic images and SR images are better. Although the bicubic images cannot provide additional resolution information for the LCC, the upsampling process can reduce some adverse effects of downsampling process in the HRNet on the LCC. However, from Table I, inputting bicubic images and SR images into HRNet greatly increase the computational cost in terms of training times and testing times.

D. Comparisons With Other Methods

It is expected that the proposed SRGDN takes advantage of the detailed structures of high-resolution feature maps in the SR branch to guide the LCC branch to achieve better performance. The HRNet is adopted as the backbone to

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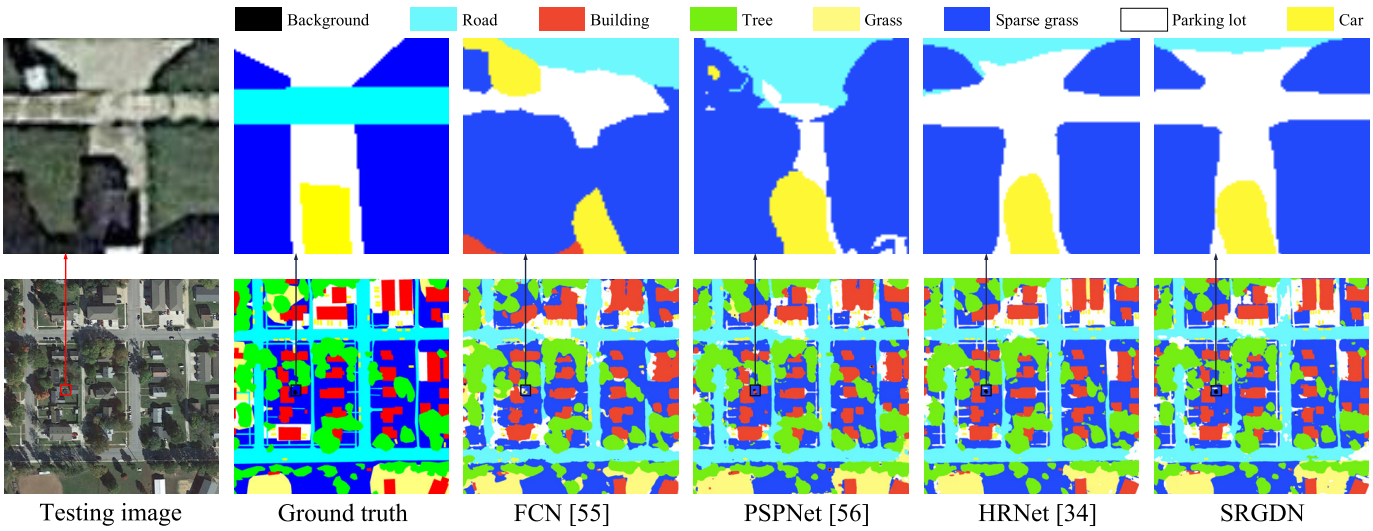


Fig. 6. Results comparison in the UAV image dataset.

preserve much resolution information, which is beneficial to the LCC performance. The HRNet is compared with the SRGDN to show the effectiveness of the proposed method. Two other methods, i.e., FCN [55] and pyramid scene parsing network (PSPNet) [56], are considered to demonstrate that the resolution information plays an important role in the LCC. In [55], it is the first time to propose encoding-decoding (i.e., downsampling–upsampling) architecture to capture both local and global information for the LCC, and it is also the first pixel-to-pixel method based on CNN to classify land cover. In [56], the pyramid pooling module is added after the backbone to make feature maps with different scales and fuse these feature maps, where the multiscale fusion strategy is fully applied for the LCC and the entire model is simplified. All results are shown in Tables I–III. The OA, CA, mIoU, and IoU are calculated from different methods and different datasets to verify the superiority of the proposed SRGDN over the other methods.

1) Experiment 1. UAV Image Dataset: First, we perform an experiment on UAV image dataset. Table I shows the OA, CA, mIoU, and IoU of six different methods. In this case, the proposed SRGDN method obtains better results than those results from the HRNet, the PSPNet, and the FCN, even inputting bicubic images into HRNet. This demonstrates that the SR branch helps the LCC branch to improve its performance. It can also be observed that the performances yielded by the HRNet are better than those from the FCN and the PSPNet, which confirms that the resolution information has a great effect on the LCC. Moreover, from Table I, the SRGDN greatly improves computational efficiency according to training times and testing times, compared with that directly inputs bicubic images and SR images into HRNet. Since the SR branch and the guidance module are removed in the testing period, the testing time of the HRNet and the SRGDN are the same. Fig. 6 shows the LCC maps of the testing image provided by all four methods (i.e., the FCN, the PSPNet, the HRNet, and the SRGDN).

TABLE II

LCC RESULTS BY THE FCN [55], PSPNET [56], HRNET [34], AND SRGDN ON THE GAOFEN-2 SATELLITE IMAGE DATASET

Methods	FCN	PSPNet	HRNet	SRGDN
training times	33538 seconds	10037 seconds	5154 seconds	12649 seconds
testing times	3332 seconds	4455 seconds	681 seconds	681 seconds
OA	74.36	78.85	81.14	82.09
CA of Background	63.11	81.27	84.71	74.51
CA of Water	64.40	55.57	77.31	69.46
CA of Vegetation	90.57	77.94	56.83	82.01
CA of Buiding	66.52	78.37	83.51	86.12
mIoU	55.95	60.01	62.91	65.31
IoU of Background	50.29	63.91	68.39	65.86
IoU of Water	57.44	51.41	66.48	66.56
IoU of Vegetation	70.46	60.54	51.56	62.16
IoU of Buiding	45.60	64.18	65.21	66.64

TABLE III

LCC RESULTS BY THE FCN [55], PSPNET [56], HRNET [34], AND SRGDN ON THE POTSDAM DATASET

Methods	FCN	PSPNet	HRNet	SRGDN
training times	404867 seconds	92303 seconds	138475 seconds	279922 seconds
testing times	9267 seconds	11462 seconds	1674 seconds	1674 seconds
OA	88.02	91.39	91.94	92.04
CA of Impervious surface	90.84	93.97	92.23	93.28
CA of Building	95.59	96.84	97.47	97.19
CA of Low vegetation	84.10	87.35	86.40	87.68
CA of Tree	84.75	82.67	88.62	86.08
CA of Car	84.95	87.33	91.21	92.19
mIoU	79.53	82.84	83.71	84.42
IoU of Impervious surface	83.01	86.46	86.55	87.44
IoU of building	88.96	93.90	93.62	94.32
IoU of Low vegetation	72.27	75.86	76.11	76.71
IoU of Tree	72.07	74.69	76.16	76.58
IoU of Car	81.34	83.28	86.10	87.05

2) Experiment 2. Gaofen-2 Satellite Image Dataset: The second experiment is performed on the Gaofen-2 satellite image dataset. The OA, CA, mIoU, and IoU of four methods

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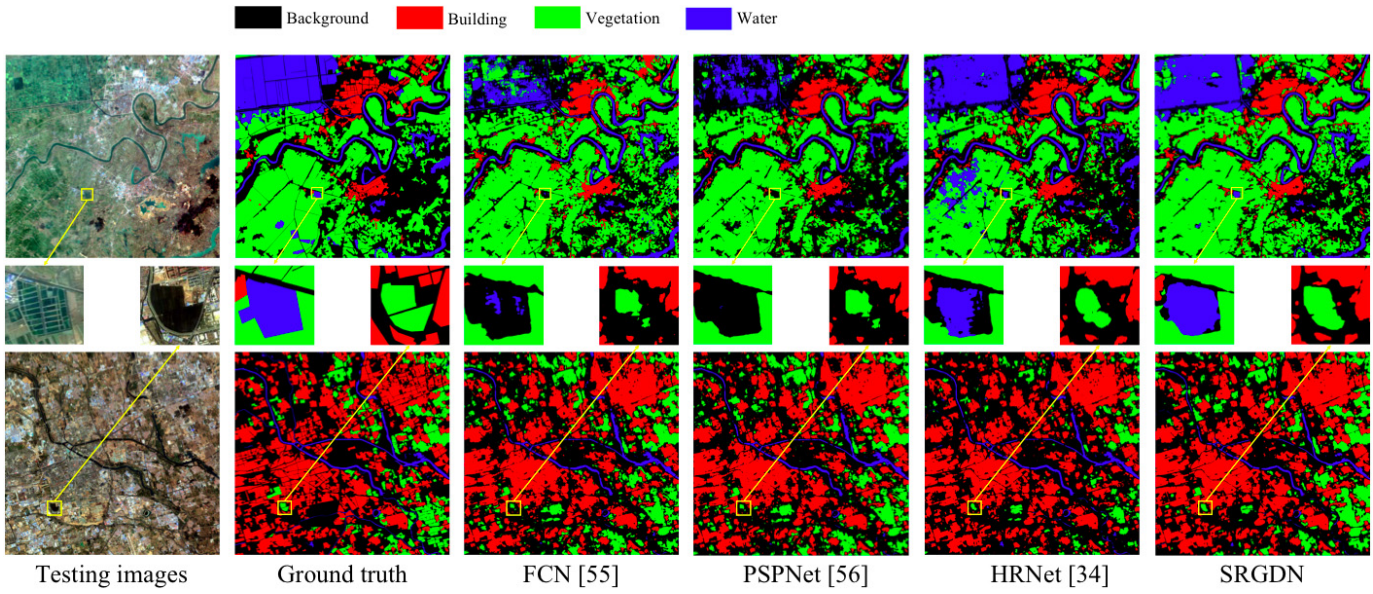


Fig. 7. Results comparison in the Gaofen-2 satellite image dataset.

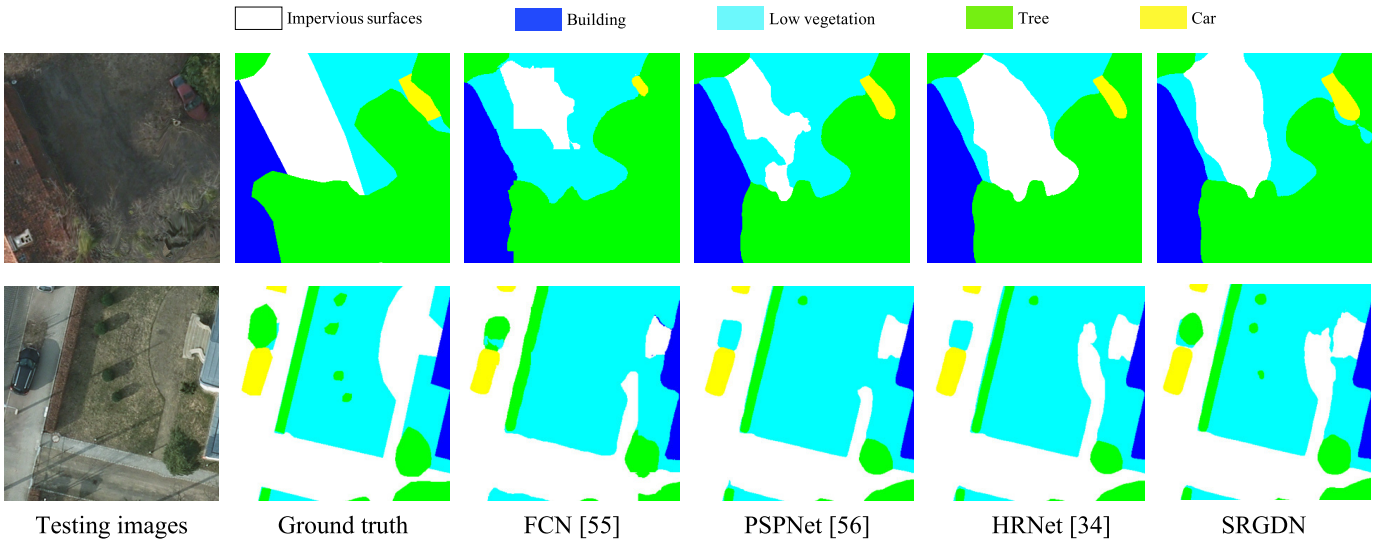


Fig. 8. Results comparison in the Potsdam dataset.

are shown in Table II. It can be observed that the proposed SRGDN achieves the highest LCC performance in terms of both OA and mIoU, which proves the effectiveness of the proposed SRGDN on the satellite image dataset. Moreover, because of retaining more resolution information, the performance of the HRNet is better than the FCN and the PSPNet. In the countryside, the vegetation area, the water area, and the background are easy to be misclassified from Fig. 7. From the perspective of the CA and the IoU of the background, the water and the vegetation, the CA of the water and the background are much higher than the IoU of those and the CA of the vegetation is slightly higher than the IoU the vegetation. Thus, the water and the background are usually oversegmented by CNN-based methods, but the proposed SRGDN can effectively extract fine edges and textures. Because the SRGDN is equivalent to the HRNet in the testing phase, and the testing times of these two methods are the same. The LCC maps of testing images obtained by four methods can be shown in Fig. 7.

3) Experiment 3. Potsdam Dataset: The third experiment is performed on the Potsdam dataset. Table III shows the OA, CA, mIoU, and IoU of the FCN, PSPNet, HRNet, and SRGDN. It can be observed that the LCC results in terms of OA and mIoU yield by the proposed SRGDN method are better than the LCC results from the other three methods. Similarly, the results of the HRNet are better than those of the PSPNet and FCN. Compared with the UAV image dataset (0.15 m per pixel) and Gaofen-2 satellite image dataset (4 m per pixel), due to the very high resolution of the images in the Potsdam dataset (0.05 m per pixel), improving resolution

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for this dataset cannot greatly improve LCC results. However, some small and unclear targets of the interesting classes can be better identified by the proposed method. For example, Fig. 8 shows some LCC patches from testing images, i.e., the low vegetation near the car in the first image patches and the tree surrounded by low vegetation in the second image patches, can be better recognized by the SRGDN. Furthermore, according to the CA and IoU of the car from all four methods, improving the resolution for remote sensing images can achieve a great LCC improvement.

V. Conclusion and Future Works

In this article, a novel end-to-end framework, including an SR branch, an LCC branch, and a guidance module, was designed for the LCC. Specifically, the EDSR was first taken as the backbone of the SR branch to obtain finer information for remote sensing images. Then, the LCC branch took the HRNet as the backbone, which could keep more resolution information. Furthermore, in the guidance module, the feature maps from the SR branch guided the LCC branch to generate the feature maps with more detailed structures based on the cosine distance and $l1$ distance. In addition, the SPP was utilized to match the size of these feature maps from different branches. Finally, a GAN was used to produce the downscaling kernels and the high- and low-resolution image pairs for training the SR branch are generated by these kernels. Compared with that directly input remote sensing images from the EDSR into the HRNet, the proposed SRGDN framework efficiently reduced training and testing times. Our experimental results conducted on both satellite and UAV image datasets proved the importance of resolution information for the LCC and the superiority of the proposed SRGDN over other methods.

However, in our method, the high- and low-resolution image pairs were obtained by a blind estimation method. This method relied heavily on the original image itself. For many satellite datasets, two remote sensing images, including a high spatial resolution and low spectral resolution image and a high spectral resolution and low spatial resolution image, could be provided at the same time. These two images can reflect a good mapping relationship between the high- and low-resolution images, which was expected to meet the needs of the SR task. In addition, how to predict the precise edge of the classification maps is a huge challenge for the CNN-based LCC methods. More fancy edge-oriented loss function was supposed to be designed for the LCC task to enhance the performance of the CNN-based methods. Moreover, one of our ongoing works is to introduce a more general model to classify land cover from both satellite and UAV image datasets, simultaneously.

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